Functional Specification — **VeriSocial**

CSC1097 - Project Functional Specification Dublin City University

Project Title: VeriSocial

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Introduction Section 1

Overview 1.1

Our project, VeriSocial, aims to create a platform designed to prioritise authentic human interaction and foster a community free from AI content and automated bots. Over the years, as AI technology has advanced, we see more examples of our online platforms being plagued by AI generated media / content as well as bot activity. We aim to address both these issues by incorporating a system that detects and filters out AI Content and removes Bots from the platform, creating a more trustworthy online community.

Alongside our AI filtering system, we want to encourage users in AI awareness by incorporating game-like features into our platform. Users will be able to encounter event "Activities" as they browse through our platform, deliberating testing and improving our user's AI awareness skills. Incentives will be created to promote user engagement and also incorporate a reputation system so that users will know who to trust.

Business Context 1.2

VeriSocial is planned to be deployed using Amazon Web Services (AWS). Though primarily an independent platform, there are future opportunities / paths that our project could take after development.

- **Future Platform Integration:** Businesses in the future who are interested in our model of a social platform can use our ideas to create a more refined version of our project.

- **Advertisements:** Although not planned to be added, our platform can serve as a host for potential businesses who wish to advertise their products / services. (Similar to how many modern day social media platforms advertise third party businesses.) This can include banner ads, for example, in user feeds.

Glossary 1.3

Profile:
A unique collection of settings and information regarding a user.

Frontend:
The visual aspects / the interface of the website that the users will be able to see.

Backend:
Consists of the programming / components of the website that allow it to operate and is not able to be seen by the user.

Database:
A collection of structured information or data - usually stored electronically via servers

Servers:
A computer or software system that provides data, resources or services to other computers known as clients. These work over a network to function.

SQL & MySQL:
SQL stands for "Structured Query Language". It is the language used to store, retrieve and manipulate data in relational databases.

HTML:
Stands for HyperText Markup Language. It is the standard markup language for creating web pages.

CSS:
Stands for Cascading Style Sheets. It describes how elements in HTML are displayed,

React:
A Javascript library used for frontend development.

API:
Stands for Application Programming Interface. They are the mechanisms that enable two software components to communicate to each other.

General Description Section 2

Product / System Functions 2.1

VeriSocial will be developed as a social media platform, with heavy emphasis on the theme of Anti-AI. The platform will consist of general social media features found in most everyday social apps (such as Twitter and Instagram). Users will be able to access an online web platform This will act as a base for implementing our Anti-AI and Bot system. Below are the primary functions of the VeriSocial Project.

Account Creation and Management
Users will be able to create accounts along with being able to manage their accounts with secure login and logout functionality. Account Customization will be available to allow users to personalise their accounts or profiles.

Content Creation and Sharing
Users will be able to create and share content for others to see. This includes text and images similar to most modern day social media applications. Users will also be able to interact with posted / shared content via features such as liking, commenting, sharing and reporting content.

AI Content Detection
Once users upload content to the platform for others to see, it will be processed by machine learning models to scan and assess if the content appears to be AI-generated. Users will be notified if their post is suspected to be AI-generated.

Bot Detection
VeriSocial will also detect bot-like behaviours based on unusual patterns that accounts display. This can include irregular posting frequency, unusual engagement and spamming. Accounts that are suspected of Bot activity will be warned and those identified as bots will be restricted from accessing the platform.

Game Features
The platform will occasionally prompt users in an optional game event prompt. These prompts can be encountered as they are scrolling through their feeds. The prompts will challenge the user and test their AI awareness skills asking a “Is this AI?” game. Users will be incentivised to participate in these events through rewards and a level up system. Users who actively participate in these events can gain badges and levels and build reputation for their account.

Reporting and Appealing
Users can report posted content if they suspect it to be AI-generated content. Users who observe accounts with bot-like behaviours can also report them for bot-like activity. Appeals will be created for cases of false negative reports.

User Characteristics and Objectives 2.2

General Social App Users (Standard Users)

These are your typical everyday social media app users who wish to connect with others in an online space. This group consists of varying characteristics including a large age range with some being more familiar with technology than others. For VeriSocial's main goal of creating a community, all users of all kinds and backgrounds should be taken into consideration.

- In order to take into account all varying kinds of standard users the platform must implement an interface that is easy to understand and familiar with popular social media platform formats. This includes distinct / recognisable icons and responsive UI.

Moderators and Community Managers

These users will be responsible for overseeing flagged content and appeals. Their responsibilities include the manual review of cases for false negatives / positives. Moderators would need to have a moderate to high level of technical skill and familiarity with how VeriSocial works.

- Moderators / Managers should have extended access to features outside of standard users. This can include their own separate interface to manage appeals and reports made by users.

Administrators

These will be the technical experts that are responsible for maintaining the platform and ensuring all features of VeriSocial function correctly.

Operational Scenarios 2.3

The following are use case tables for our project. Below we have listed some examples of the core / main use cases that will be implemented.

Use Case ID	**USR001**
Title	User makes a post
Actors	General user
Preconditions	User is logged in and has an active class profile.
Flow of events	1. User navigates to the "Post" button. 2. User writes a message or uploads a photo or both. 3. User clicks the "Submit Post" button. 4. System performs AI Content check in the background 5. If AI content is detected the system notifies the user trying to make a post.
Postconditions	Post is published on the platform.

Use Case ID	USR002
Title	User encounters “Identify AI” game prompt in feed
Actors	General user
Preconditions	User is scrolling through their feed
Flow of events	1. User encounters a post with an “Is this AI?” prompt. 2. User clicks the prompt and selects an answer. 3. If the user’s answer is correct, they earn points or a badge.
Postconditions	Points or a badge are awarded to the user if the response is correct.

Use Case ID	USR003
Title	Report suspicious bot activity
Actors	General user, Moderator
Preconditions	User has identified an account that exhibits bot-like behaviour
Flow of events	1. User navigates to the suspicious account. 2. User clicks “Report Bot Activity.” 3. System records the report and flags the account for review by a moderator. 4. Moderator reviews and confirms / rejects the bot status.
Postconditions	The account is flagged as a bot if confirmed, resulting in restriction or suspension.

Use Case ID	USR004
Title	Review and manage reports
Actors	Moderator
Preconditions	Users report AI-generated content or suspicious bot activity.
Flow of events	1. System notifies the moderator of a new report. 2. Moderator reviews reported content or account. 3. Moderator confirms or dismisses AI-generated flag or bot report. 4. Action taken on flagged content/account if necessary.
Postconditions	Reported content or accounts are addressed appropriately (confirmed as AI / bot or cleared).

Constraints 2.4

Below are some general constraints / obstacles that we expect to encounter during the project development process.

- **Content Verification Speed**: Real-time verification must be implemented. One problem we can face is the speed of verification when users are uploading content. In order to create a reliable platform, the AI verification must be implemented without noticeable delays for better user experience.

- **False Positives and Negatives**: Working with an AI detection system, there inevitably will be cases where the system will deem something incorrectly. The system must be accurate to avoid wrongful flagging or missed detections. One big challenge is the ability to work around getting false positives and negatives. Currently an appeal system will be implemented so that admins can manually review these instances.

- **Game Familiarity and Balance**: The game events must be familiar for users so as to not cause confusion and frustration. Game event prompts must also be balanced as to not detract from the social aspect of the platform. This can be adjusted and balanced during development and after user testing.

Functional Requirements Section 3

User Post Creation and Upload 3.1

Requirements ID: FR001
Description:
The platform must allow users to create and share posts (text, images or both) within their personal feed. This feature should allow users to express themselves and be able to engage with other users in the VeriSocial platform. Text and images should be supported, giving the user more options and variety in expressing themselves. Posts should also be interactable, e.g. liking, commenting and be reportable.
Criticality:
High - This feature is essential as it forms the core of user interactions within the platform. Being able to post content is a core functionality found in all social media apps, without it users would not be able to socialise with each other which defeats the purpose of our project.
Technical Issues:

- Requires backend support for storing content.
- Implementing AI analysis during upload may take time to process.

Dependencies:

- Relies on the AI detection requirement (FR002) for content scanning.
- User Profiles are required to associate posts with user accounts.

AI Detection for Posts 3.2

Requirements ID: FR002
Description:
The system must include an AI-detection feature to analyse and flag potentially AI-generated content during upload time. This functionality serves as our primary feature to emphasis and preserve our theme of human-authentic content. The AI-detection should work on text and image posts and be able to flag instances of suspected AI-content.
Criticality:
High - The AI-detection is a core feature that differentiates VeriSocial from other social platforms. It is one of our main features for driving a human-authentic online environment.
Technical Issues:

- High Detection accuracy is needed to minimise false positives (flagging human content as AI) and false negatives (allowing AI-content to be posted).
- Processing times should be as low as possible to avoid delayed uploading times.

Dependencies:

- Relies on FR001 as this feature runs automatically during post uploading.

User Login & Logout 3.3

Requirements ID: FR003
Description:
The platform must provide a secure user login and authentication system that enables users to create accounts and login with their own credentials. The login system should support standard email / password login. Once users are logged in, they should be able to log out whenever they want.
Criticality:
High - Users must be able to login as it is essential for all aspects of user interactions on the platform. Without it, we can not maintain personalised user experiences and other functionalities that are tied to the individual user. (E.g. Posting).
Technical Issues:

- Credentials must be securely handled both during storage and during transmission.

Dependencies:

- FR001 (Posting). Only logged in (authenticated) users should be able to post content.

Bot Detection and Prevention 3.4

Requirements ID: FR004
Description:
Another important feature of VeriSocial is the bot-detection and prevention system to monitor accounts for bot-like behaviour. The system should identify suspicious activity / patterns displayed by a user. This can include for example: high-frequency posting and spamming comments / posts. Users who display bot-like behaviour will be flagged and the system will go through the account. If the system confirms that the account is likely a bot, a restriction will be automatically applied to the account. Restricted accounts would not be able to interact with the platform. Suspending accounts can also be part of penalties applied.
Criticality:
High - Another key feature of VeriSocial. This feature, alongside our AI-detection, will promote our online environment of human-authenticity.
Technical Issues:

- False Positives in bot-detection may occur. We must ensure that we accurately differentiate between genuine users and bots.

Dependencies:

- FR005 (Reporting System). This feature should handle reports submitted by users regarding suspected bot activity.

Reporting System 3.5

Requirements ID: FR005
Description:
A reporting system should be included to allow users to report suspicious bot activity and also for potentially AI-generated material. This feature should enable users to easily flag / report posts, comments, and or profiles for review. Once a report is submitted, it should be processed by a moderation team and the system's automated review system.
Criticality:
High - The reporting system will allow us to maintain our trustworthy platform with the aid of the VeriSocial community. It will empower users and engage them to uphold the human-authentic platform whilst also acting as an additional layer of verification for the AI detection and bot prevention.
Technical Issues:

- Develop an effective moderation workflow for handling reports. Automated reviewing must also be accurate to avoid errors and unnecessary restrictions on genuine users.

Dependencies:

- FR002 (AI-detection). Reports should be relayed back to the AI detector which can provide another layer of verification.
- FR004 (Bot-detection). Reports related to bot-like behaviours should be relayed to the bot-detection feature to conduct an automated review of the reported account and provide details to aid in the detection.

Game Event Prompts in User Feeds 3.6

Requirements ID: FR006
Description:
The platform should include an interactive event prompt feature that engages users periodically with a challenge as they scroll through their feeds. These prompts will present the user with an "Is this AI?" challenge which aims to stimulate their AI awareness skills. The prompt will present the user with a choice between 2 pieces of content (e.g. Images), one being AI-generated and the other being authentic. Users will respond to the prompt by selecting one of the options they deem to be the human / authentic content. Correct answers will reward the user with points and / or badges.
Criticality:
Medium - Although this is a feature which adds identity to our platform, it is not essential for the platform to function properly. This feature promotes user engagement through game-like aspects. The rewards create an incentive for users to actively participate and enhance their awareness of AI content.
Technical Issues:

- For content selection, the system must accurately select randomised AI content and Human content for the user whenever they encounter a game event prompt.

Dependencies:

- FR003 (User). Users' profile should reflect their gained points from winning these game events.

Game Rewards and Achievements 3.7

Requirements ID: FR007
Description:
The game system should reward users who actively participate and engage with the platform via correctly reporting AI-generated content or suspicious accounts and participating in the Event Prompts (FR006). The system should display the users' achievements on their profiles. Badges can also be earned through various specific actions (such as milestones being reached) and also be able to be displayed on their profiles.
Criticality:
Medium - This feature creates an incentive for users to engage with the platform and promote the theme of anti-AI but not core to the platform itself.
Technical Issues:

- Tracking and storing the data of user's points and badges.

Dependencies:

- FR003 (User). Users should have their points and badges on display when rewarded.

Notifications System 3.8

Requirements ID: FR008
Description:
The platform must include a notification system that delivers real time notifications to users regarding account activity. This can include the likes for achievements, reporting posts / accounts, updates on the platform and profile related updates (e.g. new followers). Notifications should be displayed in a clear format and must be easy to manage.
Criticality:
High - The notification system is vital for user engagement and delivering important information. This can include report responses and game notifications for e.g. achievements.
Technical Issues:

- Real-Time and Asynchronous Delivery. The notifications must be delivered promptly to the user.

Dependencies:

- FR005 (Reporting System). Users who report content should be notified about the status of their reports to encourage transparency for the user.
- FR007 (Game Rewards). Achievement-based notifications, such as reaching milestones or getting achievements.

System Architecture Section 4

The system architecture for the proposed project is to utilise a mix of microservices. This system is supposed to encompass the different requirements. The architecture is broken down into an architectural overview diagram (AOD).

Architectural Overview Diagram (AOD) 4.1

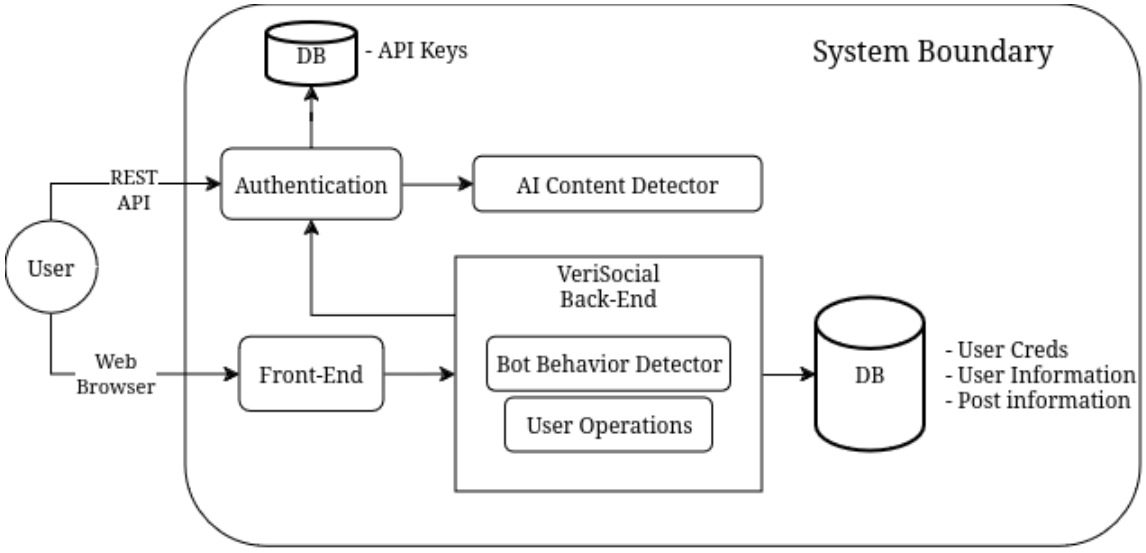


Fig.4.1 - VeriSocial Architecture Overview Diagram

The above illustrates the general architecture for the project. The project should accommodate users in the browser and potential third-party users (such as X, or Facebook) so they are able to use a limited part of our services. The architecture is supposed to be deployed in a cloud service system. The proposed services are as follows

- **Bot Behaviour Detector:** Our detection for bot behaviour will mainly be based on monitoring the data sent by the API calls. We will implement similar and more primitive techniques / concepts used in past proven technologies for bot-detection [1] [2]. This module is reliant on retrieving statistics from the database, in order to check for certain rules applied by an admin. For example, when a user posts something; this module will check their last *X* posts and see if the time difference between each post is smaller or greater than an amount specified by the admin.

- **AI Content Detector:** This module will host the AI models from **Hugging Face** [3] that will be running on an **ONNX** [4] environment. The purpose of this module is to receive queries on whether certain content is AI generated, and return both certainty and prediction of the query. It is intended for this module to receive requests from both inside, and outside our system. This service is potentially the most expensive in terms of performance, and potential costs.

- **Authentication:** The authentication is there to provide tokens for third parties trying to use the AI content detector.

- **Back-end:** The default backend for the project, and where the user operations will be maintained. Bot behaviour detection will also have to be a part of this service as this feature can only happen within the VeriSocial app. This would also take care of the user authentication within the application. This module would have an already made API key for authentication, so it's able to communicate with the AI-content detection module. This system will be built using **Rust (Actix-Web)**.

- **Front-end:** This serves as the presentation layer for the users who wish to use VeriSocial. Note that, not all expected use will come from normal users so not all users need to use the front-end. This will be built using **React and Redux** [5-9].

- **Database:** For this project, the database of choice will have to be a **relational** database, that allows us to perform different operations on the data being retrieved. This is originally planned to be a **MySQL** database.

High-Level Design Section 5

The services that the system provides need to communicate with each other to ensure good communication and efficiency. The proposed design is shown by a context diagram, and below there will be a sequence diagram showing the usual use for a user trying to post on the web app.

Context Diagram 5.1

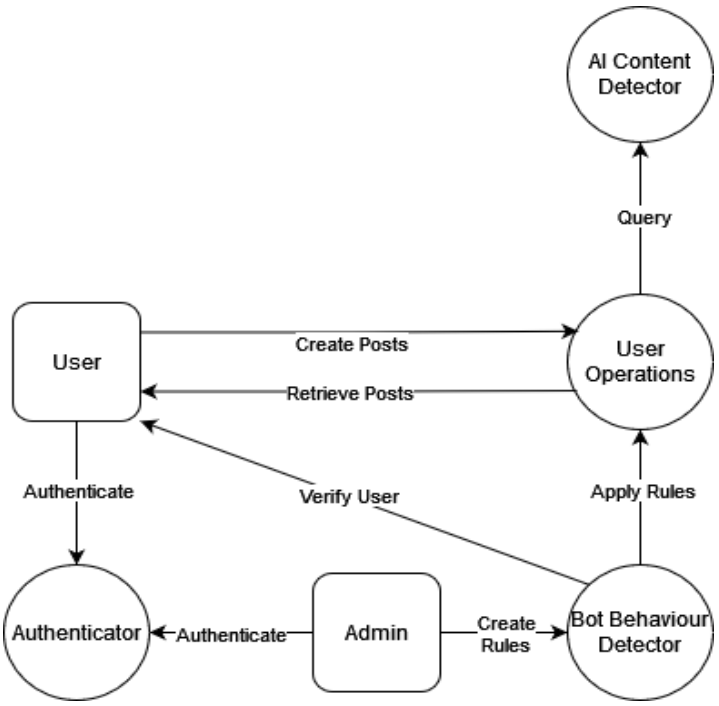


Fig.5.1 - VeriSocial Context Diagram

As seen by the diagram above the user should be able to create posts which are then posted on the timeline for other users to see.

- **User:** The main actor interacting with the system. It's able to create information for our system such as posts, user details and interact with other users.
- **Bot Behaviour:** Monitors the different posts/ interactions for signs of automation and bot-like behaviour. It receives rules made by an admin.
- **AI Content Detector:** User operations queries this module in order to analyse the different posts made by users of the system.
- **User Operations:** User operations is in charge of posting, updating user details, and the interaction between users. Thus, it is important for it to be able to communicate with the bot behaviour detector and the AI content detection system, as those modules are involved with the user.
- **Authenticator:** In charge of verifying the user and admin, ensuring they are verified before accessing the system features.

Sequence Diagram 5.2

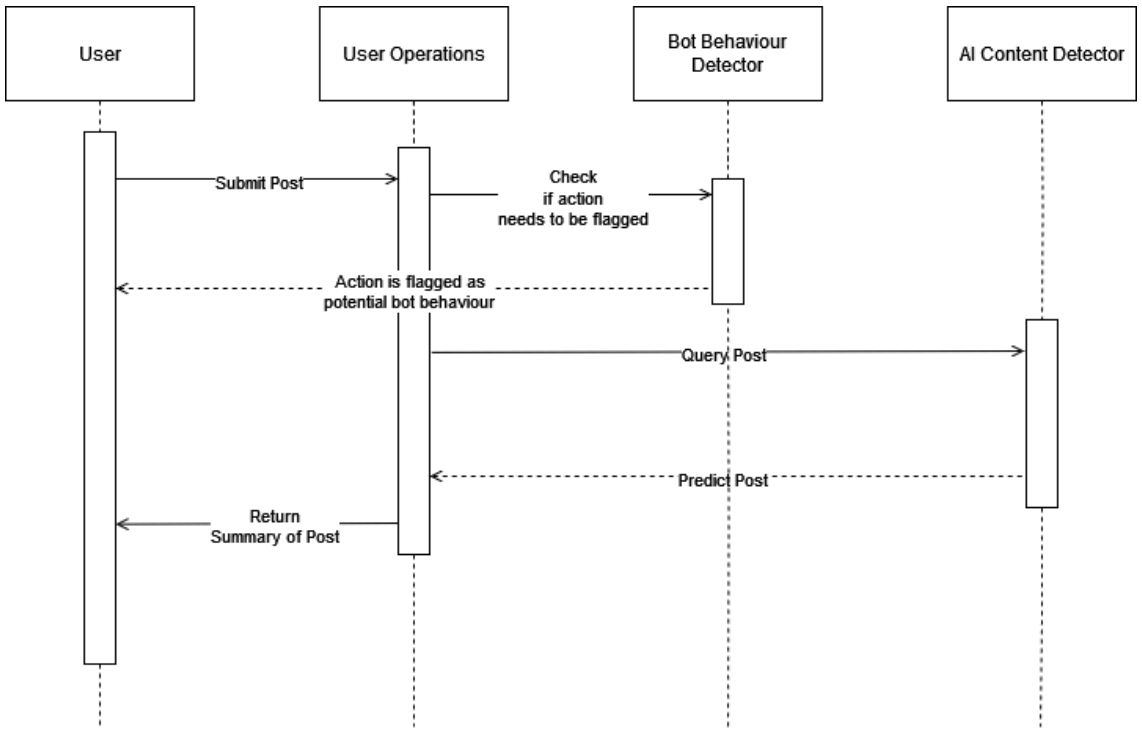


Fig.5.2 - VeriSocial Sequence Diagram

The diagram above shows the flow of a user submitting a post to our system;

- User submits posts through the UI, and this in turn lets the user operation module start verifying the authenticity of the post.

- User operations apply the rules given by the bot behaviour detector, to check if the action needs to be flagged.

    - If the action needs to be flagged, send notification to the user telling them that they are exhibiting potential bot behaviour.

- User operation also queries AI-content detectors, and the content detector predicts whether the post is AI-generated or not.

    - Note that, this process may not be done by the time the user submits the post and thus the time between the user receiving a summary of their post and receiving the actual prediction may be different.

Preliminary Schedule

Section 6

The following is a Gantt Chart depicting our initial planned development timeline for VeriSocial. It contains a high-level overview of the main sections needed to complete this project with each section having a number of features / components to be implemented. Development of the project will be organised using Jira [10] and the Gantt chart below was also created in Jira.



Fig.6.1 - Gantt Chart created in Jira

Our project is broken down into four main components as seen above. (Note that, the AI-Content Detector Service includes the authentication API for third parties). This is how workload will be broken down;

- **Javier -** Will take care of the backend and creating the AI content service, and deploying the application.
- **Joaquin -** Will take care of the front-end and the bot behaviour detector.

Note that consistent testing per module will be done proactively throughout the project.

Appendices Section 7

Below are resources used for researching our project premise as well as links to the technologies we will be using.

[1] Y. Al-Hammadi. "*Behavioural Correlation for Malicious Bot Detection*". University of Nottingham. April 2010. [Resource].

[2] K. Hayawi et al. "*Social media bot detection with deep learning methods: a systematic review*". Neural Computing and Applications. July 2020. [Resource].

[3] Hugging Face. [Resource].

[4] ONXX Documentation. [Resource].

[5] ReactJS Documentation. [Resource].

[6] Axios Library Documentation. [Resource].

[7] React Jest Documentation. [Resource].

[8] React Redux Documentation. [Resource].

[9] React Redux Saga Documentation. [Resource].

[10] Jira Software Development. [Resource].

[11] Draw.io for Diagrams and Architecture Planning. [Resource].